# SCURRY 1.2
# Message Exchange Reference

Revision 2026-10-23 | ScurryMsg source release 1.10.0 | Human-readable ASCII projections

This reference covers the currently implemented exchange families and every central endpoint/filter expansion. Arbitrary message text, recipients, attachment combinations and folder names are parameterized, not an infinite list. It does not introduce endpoints or promote the separate 1.1 resource proposal.

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# 1. Header types, names and possible values

The header catalog appears first. Every SCURRY frame begins with Protocol as its first header record. HTTP-style Name: value lines below are a readable projection of fixed-slot framing over TLS/TCP, not literal HTTP or a replacement wire format. UTF-data-* denotes UTF-8 text; ASCII-data-* is ASCII-compatible text; Binary-data-* denotes raw bytes. All sample text is ASCII to keep the projections readable.

## 1.1 Physical framing and type rules

| Part / type | Encoding and bounds |
|---|---|
| Bootstrap | 64 ASCII bytes, NUL right-padded: SCURRY 1.2 UTF-8 COUNT HEADER-BYTES PAYLOAD-BYTES. Header bytes exclude bootstrap; payload bytes exclude all headers. |
| Header record | 64-byte ASCII name, 8 ASCII decimal capacity, 8 ASCII decimal padded encoded length, then encoded padded value. |
| Header names | [A-Za-z][A-Za-z0-9'-]{0,63}; canonical names below. Reject duplicates including case variants. Protocol must be first. |
| UTF-8 text | Capacity in Unicode scalars, NUL padding then UTF-8 encoding. No embedded NUL, CR or LF. Encoded length includes padding. |
| ASCII subtype | Uses the same UTF-8 text slot; enum/number/hash restrictions additionally apply. Integers have no sign, whitespace or digit grouping. |
| Binary subtype | Only four named fields use raw 256-byte slots; capacity=encoded length=256. May contain every byte, including NUL and CR/LF. Not Base64, hex or UTF-8 on wire. |
| Limits | 1..128 records including Header-Count; <=150000 header bytes. Frame payload length 0..9223372036854775807, further limited by the host. Current native central defaults request 4 MiB/response 8 MiB; configured maxima apply. |
| Large text slots | 256 scalars: Protocol, Subject, Message-Identifier, Retrieve-Message-Identifier, sender/receiver passwords, Sender-Host, Recipient-Host, Sender-Account, Additional-Recipients, Additional-Recipients-For-Message and numbered attachment descriptors. Other text fields: 64 scalars. |
| Text projection | Blank line separates projected headers from projected body. Original wire boundaries come from lengths, never a blank-line scan. [angle placeholders] are explanatory, not bytes. Physical bootstrap and padding are omitted from exchanges. |

## 1.2 Complete known wire-header catalog

Required fields depend on endpoint and whether the frame is an outer command, nested original, response or stored metadata. A catalog entry is not permission to send every header on every operation. Header capacity is a framing bound; individual handlers may impose smaller semantic limits.

| Header | Type | Capacity | Values / use |
|---|---|---|---|
| Protocol | UTF-8 enum/path | 256 | Required first header record. SCURRY 1.2 UTF-8 UTF-8 followed by space and an enabled /endpoint. |
| Header-Count | ASCII decimal | 64 | 1..128; counts itself and every header record. Filled by codec. |
| Operating-Mode | ASCII enum | 64 | One-To-One or Central-Server. Explicit listener mode; not recipient count. |

| Header | Type | Capacity | Values / use |
|---|---|---|---|
| Sender's-Login | UTF-8 credential | 64 | Account/contact login; scoped delegated login for remote notice, retrieval and token receipt. Nonempty. |
| Sender's-Password | UTF-8 credential | 256 | Central: direct configured account password; delegated: 64 hex characters. Peer: 64 printable ASCII password + " ~;~ " + fresh 16-digit nonce + " ~;~ " + UTC epoch. |
| Receiver's-Login | UTF-8 credential | 64 | Receiving account login. Authorizes only /message-retrieval-from-sender-for-storage. |
| Receiver's-Password | UTF-8 credential | 256 | Receiving account password for final import only; sender credentials do not authorize import. |
| Sender-Full-Name | UTF-8 identity | 64 | Original authenticated full name, e.g. Alice Example; no CR/LF/NUL. |
| Sender-Email | UTF-8 identity | 64 | Original authenticated email; server replaces client labels on ordinary central send. |
| Recipient-Name | UTF-8 identity | 64 | Exact recipient full name; ordinary central exact selector may leave name empty. |
| Recipient-Email | UTF-8 identity | 64 | Recipient email, e.g. bob@example.net; immutable binding in token flow. |
| Subject | UTF-8 text | 256 | Original subject, at most 256 scalars; Example subject. |
| Message-Identifier | UTF-8 structured ID | 256 | Conversation ID: sanitized original full name, email, MD5(email), MD5(subject), hyphenated UUID, original UTC epoch; separator " ~;~ ". Preserve on replies. |
| Delivery-Identifier | ASCII ID | 64 | 32 lowercase hexadecimal UUID characters; one delivery correlation scope. Fresh application replies use a fresh value. |
| Retrieve-Message-Identifier | UTF-8 structured ID | 256 | Original conversation ID requested by recipient-history operation. |
| Request-Identifier | ASCII ID | 64 | Optional stable 32 lowercase hex send idempotency key. Same account/key/content returns prior send result; conflicting content rejected. |
| Server-Message-Id | ASCII ID | 64 | 32 lowercase hex server-owned mailbox copy ID; never an authorization credential. |
| Notice-Id | ASCII ID | 64 | Original receiver mailbox notice ID, retained when replaced by imported payload. |
| Sender-Account | UTF-8 server metadata | 256 | Server-owned sender account key. Never trust a client-supplied account label as authorization. |
| Localization | UTF-8 text | 64 | Configured locale, e.g. en-US; informational. |
| Country-Of-Origin | UTF-8 text | 64 | Configured country, e.g. IN; informational, may be empty. |
| Time-Zone | UTF-8 text | 64 | Configured zone, e.g. Asia/Kolkata or UTC; informational. |
| Send-Time | ASCII decimal | 64 | Fresh per-attempt UTC Unix seconds; changes on retries for authentication freshness. |

| Header | Type | Capacity | Values / use |
|---|---|---|---|
| Message-Sent-Time | ASCII decimal | 64 | Frozen original nonnegative UTC Unix seconds. Receipt implementation accepts at most 12 decimal digits. Preserve across resume/retries. |
| Random | ASCII random text | 64 | Fresh 16 ASCII characters; clients use uppercase hex. Distinct from the binary Random-Id and Random-Password. |
| One-To-One | ASCII flag | 64 | 1 for a single-recipient composition; 0 for a multiple-recipient composition. Not Operating-Mode. |
| Body-Length | ASCII byte count | 64 | Original primary body byte length for content/reference metadata; receipt envelope uses receipt body length. Notice/chunk frame length can differ. |
| Body-Format | ASCII media type | 64 | text/plain, text/html, text/rtf; application/json for sender-validation manifests. |
| Total-Length | ASCII byte count | 64 | Full original plaintext body plus all attachment bytes; nonnegative signed-64-bit maximum, subject to host limits. |
| Content-SHA256 | ASCII digest | 64 | 64 lowercase hex characters over full original payload; not a signature. Not the digest of a retrieval chunk. |
| Receipt-Algorithm | ASCII enum | 64 | SCURRY-PRNG-SHA512-1 for 1.2 checksum receipts. |
| Resume | ASCII flag | 64 | 0 for initial transfer; 1 for peer resume with a nonzero saved offset. |
| Resume-Offset | ASCII byte count | 64 | Saved peer byte count, or source retrieval range start; 0..original total. |
| Resume-Tail-SHA256 | ASCII digest | 64 | 64 lowercase hex over last min(128,saved-offset) bytes; present for nonzero peer saved offset. |
| Transfer-Fingerprint | ASCII digest | 64 | 64 lowercase hex of immutable selected metadata, including frozen epoch and receipt algorithm; excludes fresh authentication fields. |
| Complete | ASCII flag | 64 | 0 incomplete, 1 durably completed. Completed peer status reply is followed by the saved receipt frame. |
| Retrieval-Length | ASCII decimal | 64 | 1..65536 requested source chunk bytes; last response may be shorter at EOF. |
| Content-Type | ASCII media type | 64 | application/octet-stream; application/json; charset=utf-8; text/plain; charset=utf-8; text/tab-separated-values; charset=utf-8; text/scurry-headers; charset=us-ascii; application/x-scurry-frames. Semicolons separate parameters, not multiple values. |
| Status-Code | ASCII status | 64 | 200 success; common host errors 400 invalid, 403 denied, 404 unavailable, 501 unsupported callback. A host may close malformed/unauthenticated connections instead. |
| Message-Count | ASCII decimal | 64 | Nonnegative count of selected records/frames/results. Empty success is 0 with empty payload. |
| Folder | UTF-8 metadata | 64 | Logical mailbox folder, e.g. ScurryMails, Sent, Work. Folder routes carry URI-encoded tokens, not filesystem paths. |

| Header | Type | Capacity | Values / use |
|---|---|---|---|
| Read | ASCII metadata | 64 | true or false in stored/listed metadata; retrieval does not inherently mark read. |
| Category | ASCII decimal | 64 | 0 uncategorized or 1..8. Independent of read/unread. |
| Message-Length | ASCII byte count | 64 | Stored message payload length; 0 for header-only notice. |
| Sender-Host | UTF-8 location | 256 | Reachable sender service hostname; identity checked with configured TLS trust. Not an HTTP URL. |
| Sender-Port | ASCII port | 64 | Configured TCP port 1..65535, e.g. 33338. |
| Recipient-Host | UTF-8 routing text | 256 | Recipient server hostname carried by routing helpers when used; optional and not a substitute for configured destination/authorization. |
| Token-Notice | ASCII flag | 64 | 1 identifies stored header-only cross-server notice; metadata is unverified until source retrieval. |
| Token-Request | RAW binary | 256 | Exactly 256 zero octets on issuance request only. |
| Random-Id | RAW binary | 256 | Exactly 256 cryptographically random octets; reuse the issued delivery value throughout that flow. |
| Random-Password | RAW binary | 256 | Exactly 256 independently random octets; distinct from the account password. |
| Sender's-Token | RAW binary | 256 | Exactly 256 server-generated random octets, bound to one archived message/delivery. |
| Delegated-Login | ASCII secret | 64 | 32 hex characters returned at issuance and in account-owned full notice retrieval. Use as Sender's-Login for scoped operations. |
| Delegated-Password | ASCII secret | 64 | 64 hex characters returned at issuance/full notice retrieval; use as Sender's-Password for scoped operations. Never forward general account password. |
| Additional-Recipients | UTF-8 descriptor | 256 | text/tab-separated-values ~;~ UTF-8 ~;~ none ~;~ SIZE ~;~ OFFSET ~;~ recipients.tsv. SIZE/OFFSET refer to outer request payload. Separate from numbered attachments. |
| Additional-Recipients-For-Message | UTF-8 TSV row | 256 | One recipient name<TAB>email without a heading or newline. Server-generated recipient copy metadata. |
| Attachment-01..Attachment-32 | UTF-8 descriptor | 256 | MIME ~;~ encoding ~;~ endian ~;~ SIZE ~;~ OFFSET ~;~ filename. Sparse indices allowed. endian: big/little/none. Arbitrary file encoding: binary; body text: UTF-8. Offsets count from original payload start. Descriptor <=256 UTF-8 bytes; shorten filename only. |

## 1.3 Receipt body fields - not wire-header records

| Body field | Type / allowed value |
|---|---|
| Checksum-Algorithm | SCURRY-PRNG-SHA512-1 |
| Message-SHA512 | 128 lowercase hexadecimal digits. Full original plaintext payload plus deterministic 8192-byte suffix; see design for exact arithmetic. |
| Sender-Name / Sender-Email | Original sender full name/email, single line. |
| Receiver-Name / Receiver-Email | Expected receiving identity, single line. |
| Sent-Time | Original frozen epoch represented in receiver-local ISO timestamp with numeric UTC offset and [zone]. |
| Received-Time | Durable completion time represented the same way. Preserve exact original strings on retries. |

A receipt contains exactly the eight LF-terminated lines shown below, in this order; strict UTF-8, maximum 16384 bytes. SHA-256 and the deterministic SHA-512 receipt are checksums, not signatures or proof a human read the message.

```
Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

## 1.4 Contact/address-book JSON field catalog

The local contact directory has its own version-1 UDP JSON protocol. These are JSON keys, not SCURRY 1.2 headers. ASCII-style projections must not add Protocol, Host or HTTP request lines to the actual UDP datagram.

| Key | Type / possible values |
|---|---|
| version | Integer 1; independent of SCURRY wire version 1.2. Omitted request defaults to 1. |
| op | ping, get, list. No remote add/update/delete. |
| request_id | Optional string <=64 characters, echoed; use fresh values and verify matching replies. |
| token | Optional shared UTF-8 token when listener configured. Non-loopback binding requires token-file (32..256 characters). Plain UDP is not encrypted. |
| name / email / city | get by exact normalized name and email; optional city disambiguates. Name/city <=150 chars; valid email <=254 chars. Email domain case normalized, local-part case preserved. |
| id | get by local contact UUID string <=36 characters; takes precedence over name/email. |
| offset / limit | list offset integer >=0; limit integer 1..10; defaults 0/10. |
| ok | Boolean true/false. |

| Key | Type / possible values |
|---|---|
| app | ping response currently reports legacy packaged name ScurryMsg.com; branding does not change protocol. |
| contact / contacts | get returns one contact object; list returns array. |
| total / next_offset | Integer contact count; next_offset integer or null at end. Pages may contain fewer rows to keep bytes bounded. |
| error | not_found, ambiguous, expired, unauthorized, bad_request. |
| message | Optional explanatory error text; bad-request exception detail truncated to 240 chars. |
| contact.id / name / email / city | Local UUID and validated identity/location text. |
| contact.address / ip_version | Public IPv4/IPv6 address and integer 4 or 6. No application TCP port is discovered. |
| contact.imported_at / expires_at | Unix seconds; expiry can be null. Unknown expiry is not assumed to be another day after import. |
| contact.expired / freshness | Boolean expired; freshness expiry_provided or unknown. |

## 1.5 Structured body and descriptor types

| Body type | Fields / allowed values |
|---|---|
| Search JSON | subject, sender, text: optional strings <=256 chars. ANDed case-insensitive substring matching. Unknown keys rejected. |
| Status JSON | {"value":"Available"}; grouped uses {"group":"Work","value":"Away"}. value <=4096 chars; group nonempty <=64. Not message headers. |
| ID selection TSV | Heading message-id, then 1..1000 distinct 32-lowercase-hex IDs. LF/CRLF accepted. get-message exactly one; delete nonempty; get-all empty means all within scope. |
| Recipient selector TSV | Heading match<TAB>name<TAB>email; match exact or wildcard. Glob *, ?, [] only in directory matching, never commands. Up to 1000 selectors/recipients. Separate outer descriptor, at most 1 MiB in reference backend. |
| Acceptance TSV | message-id<TAB>name<TAB>email. Independent recipient copies. Delivery receipt still required. |
| Deletion TSV | message-id<TAB>result; result deleted. All selected IDs must be in account and route scope. |
| ASCII header listing | Name: JSON-quoted-ASCII-string, CRLF lines and blank lines between records. Non-ASCII escaped as \uXXXX. This is a response body, distinct from outer framing. |
| Validation JSON | schema scurry.sender-authentication/1; action request/response; kind webcam/dynamic-image; challenge (64 lowercase hex), issued_utc/expires_utc (integer UTC Unix seconds), context.gps, context.system. Response adds response_utc, owner_email, sha256, sha512; dynamic responses also image_version, previous_checksums. |
| Validation context | Fresh local GPS and device/system fields are application manifest data, not extra wire headers. Exact schemas and consent rules remain in feature source/documentation. |
| ScurryLink .conf | One [ScurryLink] section: FullName, Email, Address; optional City, ExpiresAt. UTF-8 <=16 KiB. Import validates address, preserves no PIN or extra credential fields. |

## 2. How to read the exchanges

Each REQUEST/RESPONSE block is one readable frame or explicitly identified UDP datagram. Header-Count is the count of displayed wire-header records and includes itself. It is not a byte count. Long values wrap for PDF layout only. Illustrative credentials, IDs and checksums must be replaced/computed for a real run. Values such as [NESTED FRAME] label the boundaries of a serialized inner frame, not literal payload delimiters. Frame payload byte counts belong to the binary bootstrap and are intentionally not invented as HTTP Content-Length headers.

The base text example body is 29 bytes; its Content-SHA256 and receipt checksum are computed from those bytes. Variants with attachment placeholders require new actual lengths and checksums. The sample fixed epoch/nonce is for documentation only; live authentication requires fresh clocks/nonces. Normal central account passwords keep their configured format; scoped passwords and peer tuples have the formats in the catalog.

| Section | Coverage |
|---|---|
| 3 | Every peer content endpoint, capabilities, request/reply, resume and completion replay, recipient history. |
| 4 | Single central server: send, acceptance, listing, retrieval, acknowledgement, sender receipt fetch. |
| 5 | Multiple central servers: issuance, notice, notice retrieval, chunk retrieval/resume, receiver import, receipt submission/fetch. |
| 6 | Contact book: ping/get/list, pagination, IPv4/IPv6, token and all error codes; manual .conf operations and optional HTTPS download. |
| 7 | Every central endpoint plus folder/read/unread/category1..8 expansions, search, fan-out, status and errors. |
| 8 | Coverage and implementation limits; documentation packaging. |

# 3. Client-to-client exchanges

Actor A is the sending peer; actor B is the receiving peer. Default TCP 33337, authenticated verified TLS. Contact discovery is separate (section 6). Each content transfer may first perform the status query shown later. Responses contain the receiver identity and the original delivery correlation.

## 3.1 /message

A sends content to B. B returns the receipt on this transfer connection only after verifying and durably storing the full payload.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 26
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 17
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Localization: en-US
Country-Of-Origin: IN
Time-Zone: UTC
Send-Time: 1792137660
Random: FEDCBA9876543210
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 394
Content-Type: text/plain; charset=utf-8

Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

## 3.2 /mailbox-public

A sends content to B. B returns the receipt on this transfer connection only after verifying and durably storing the full payload.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /mailbox-public
Header-Count: 26
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 17
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Localization: en-US
Country-Of-Origin: IN
Time-Zone: UTC
Send-Time: 1792137660
Random: FEDCBA9876543210
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 394
Content-Type: text/plain; charset=utf-8

Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

## 3.3 /ping

A sends content to B. B returns the receipt on this transfer connection only after verifying and durably storing the full payload.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /ping
Header-Count: 26
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 17
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Localization: en-US
Country-Of-Origin: IN
Time-Zone: UTC
Send-Time: 1792137660
Random: FEDCBA9876543210
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 394
Content-Type: text/plain; charset=utf-8

Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

## 3.4 /invitation

A sends content to B. B returns the receipt on this transfer connection only after verifying and durably storing the full payload. The .ics attachment carries UTC schedule and RSVP attendees; protocol delivery is separate from accepting/declining the invitation.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /invitation
Header-Count: 27
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: [body-size]
Body-Format: text/plain
Total-Length: [body-plus-ics-size]
Content-SHA256: [64-lowercase-hex-full-payload]
One-To-One: 1
Resume: 0
Resume-Offset: 0
Attachment-01: text/calendar ~;~ binary ~;~ none ~;~ [ics-size] ~;~ [body-size] ~;~ invitation.ics

UTF-data-invitation-text
Binary-data-calendar-attachment
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 17
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Localization: en-US
Country-Of-Origin: IN
Time-Zone: UTC
Send-Time: 1792137660
Random: FEDCBA9876543210
Content-SHA256: [64-lowercase-hex-full-payload]
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: [computed-receipt-byte-count]
Content-Type: text/plain; charset=utf-8

[exact eight-line receipt for this full original payload]
```

## 3.5 /greeting-card

A sends content to B. B returns the receipt on this transfer connection only after verifying and durably storing the full payload.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /greeting-card
Header-Count: 27
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: [body-size]
Body-Format: text/plain
Total-Length: [body-plus-png-size]
Content-SHA256: [64-lowercase-hex-full-payload]
One-To-One: 1
Resume: 0
Resume-Offset: 0
Attachment-01: image/png ~;~ binary ~;~ none ~;~ [png-size] ~;~ [body-size] ~;~ greeting.png

UTF-data-greeting-text
Binary-data-rendered-PNG
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 17
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Localization: en-US
Country-Of-Origin: IN
Time-Zone: UTC
Send-Time: 1792137660
Random: FEDCBA9876543210
Content-SHA256: [64-lowercase-hex-full-payload]
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: [computed-receipt-byte-count]
Content-Type: text/plain; charset=utf-8

[exact eight-line receipt for this full original payload]
```

## 3.6 /sender-authentication

A sends content to B. B returns the receipt on this transfer connection only after verifying and durably storing the full payload. Manifest subobjects are labeled placeholders. Webcam capture is consent-based; challenges are bound, expire and are consumed on approval.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /sender-authentication
Header-Count: 26
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: [manifest-byte-length]
Body-Format: application/json
Total-Length: [manifest-byte-length]
Content-SHA256: [64-lowercase-hex-manifest]
One-To-One: 1
Resume: 0
Resume-Offset: 0

{
  "schema": "scurry.sender-authentication/1",
  "action": "request",
  "kind": "webcam",
  "challenge": "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa",
  "issued_utc": 1792137600,
  "expires_utc": 1792137900,
  "context": {
    "gps": "[fresh-local-GPS-fix-object]",
    "system": "[device-system-profile-object]"
  }
}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 17
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Localization: en-US
Country-Of-Origin: IN
Time-Zone: UTC
Send-Time: 1792137660
Random: FEDCBA9876543210
Content-SHA256: [64-lowercase-hex-manifest]
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: [computed-receipt-byte-count]
Content-Type: text/plain; charset=utf-8

[exact eight-line receipt for this full original payload]
```

### Application response for /sender-authentication

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /sender-authentication
Header-Count: 27
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Sender's-Login: ASCII-data-pair-login
Sender's-Password: BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB ~;~ 0000000000000002 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 66666666666646668666666666666666
Recipient-Name: Alice Example
Recipient-Email: alice@example.org
Subject: Validation response
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
One-To-One: 1
Body-Format: application/json
Body-Length: [manifest-size]
Total-Length: [manifest-plus-PNG-size]
Content-SHA256: [full-payload-sha256]
Attachment-01: image/png ~;~ binary ~;~ none ~;~ [png-size] ~;~ [manifest-size] ~;~ validation.png
Resume: 0
Resume-Offset: 0

{
  "schema": "scurry.sender-authentication/1",
  "action": "response",
  "kind": "webcam",
  "challenge": "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa",
  "issued_utc": 1792137600,
  "expires_utc": 1792137900,
  "response_utc": 1792137660,
  "owner_email": "bob@example.net",
  "sha256": "[PNG-SHA256]",
  "sha512": "[PNG-SHA512]",
  "context": {
    "gps": "[fresh-fix-object]",
    "system": "[device-profile-object]"
  }
}
Binary-data-consented-1920x1080-PNG
```

This is a new B-to-A content delivery with the same conversation and a fresh delivery ID. A returns its own /delivery-success receipt, with identities reversed and checksum recomputed. It is not the receipt for the preceding validation request.

## 3.7 /sender-authentication-dynamic-image

A sends content to B. B returns the receipt on this transfer connection only after verifying and durably storing the full payload. Manifest subobjects are labeled placeholders. Webcam capture is consent-based; challenges are bound, expire and are consumed on approval.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /sender-authentication-dynamic-image
Header-Count: 26
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: [manifest-byte-length]
Body-Format: application/json
Total-Length: [manifest-byte-length]
Content-SHA256: [64-lowercase-hex-manifest]
One-To-One: 1
Resume: 0
Resume-Offset: 0

{
  "schema": "scurry.sender-authentication/1",
  "action": "request",
  "kind": "dynamic-image",
  "challenge": "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa",
  "issued_utc": 1792137600,
  "expires_utc": 1792137900,
  "context": {
    "gps": "[fresh-local-GPS-fix-object]",
    "system": "[device-system-profile-object]"
  }
}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 17
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Localization: en-US
Country-Of-Origin: IN
Time-Zone: UTC
Send-Time: 1792137660
Random: FEDCBA9876543210
Content-SHA256: [64-lowercase-hex-manifest]
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: [computed-receipt-byte-count]
Content-Type: text/plain; charset=utf-8

[exact eight-line receipt for this full original payload]
```

### Application response for /sender-authentication-dynamic-image

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /sender-authentication-dynamic-image
Header-Count: 27
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Sender's-Login: ASCII-data-pair-login
Sender's-Password: BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB ~;~ 0000000000000002 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 66666666666646668666666666666666
Recipient-Name: Alice Example
Recipient-Email: alice@example.org
Subject: Validation response
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
One-To-One: 1
Body-Format: application/json
Body-Length: [manifest-size]
Total-Length: [manifest-plus-PNG-size]
Content-SHA256: [full-payload-sha256]
Attachment-01: image/png ~;~ binary ~;~ none ~;~ [png-size] ~;~ [manifest-size] ~;~ validation.png
Resume: 0
Resume-Offset: 0

{
  "schema": "scurry.sender-authentication/1",
  "action": "response",
  "kind": "dynamic-image",
  "challenge": "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa",
  "issued_utc": 1792137600,
  "expires_utc": 1792137900,
  "response_utc": 1792137660,
  "owner_email": "bob@example.net",
  "sha256": "[PNG-SHA256]",
  "sha512": "[PNG-SHA512]",
  "context": {
    "gps": "[fresh-fix-object]",
    "system": "[device-profile-object]"
  },
  "image_version": "[version]",
  "previous_checksums": "[checksum-history-array]"
}
Binary-data-consented-1920x1080-PNG
```

This is a new B-to-A content delivery with the same conversation and a fresh delivery ID. A returns its own /delivery-success receipt, with identities reversed and checksum recomputed. It is not the receipt for the preceding validation request.

## 3.8 Capability discovery

Capability list/host data is abbreviated; consult capabilities.py/schema for all informational server/host properties. It does not negotiate a version or authorize unlisted functions.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-functionality
Header-Count: 14
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-functionality
Header-Count: 8
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Content-Type: application/json; charset=utf-8

{
  "protocol_version": "1.2",
  "operating_mode": "One-To-One",
  "schema": "scurry.server-functionality/1",
  "server": {
    "id": "ASCII-data-service-id",
    "name": "Bob peer",
    "software": "ScurryMsg",
    "software_version": "1.10.0"
  },
  "generated_utc": "2026-10-23T00:00:00Z",
  "functionalities": [
    {
      "path": "/list-functionality",
      "kind": "literal",
      "authentication_required": true
    }
  ],
  "response_functionalities": [
    "/delivery-success",
    "/transfer-state",
    "/list-functionality",
    "/message-recipients-information-retrieval"
  ],
  "host": "[bounded-host-and-address-metadata]"
}
```

## 3.9 Initial status before sending

Zero payload in both frames. No saved bytes means no Resume-Tail-SHA256.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /transfer-status
Header-Count: 15
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Transfer-Fingerprint: [64-hex-immutable-fingerprint]
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /transfer-state
Header-Count: 11
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Random: 0123456789ABCDEF
Resume-Offset: 0
Complete: 0
```

## 3.10 Resume after interrupted transfer

For this separate large-payload example, let the original total be 10000 bytes. Validate tail/fingerprint, then open a fresh transfer connection with fresh auth and send the following. Body/attachment metadata and delivery IDs remain unchanged.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /transfer-status
Header-Count: 15
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Transfer-Fingerprint: [64-hex-immutable-fingerprint]
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /transfer-state
Header-Count: 12
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Random: 0123456789ABCDEF
Resume-Offset: 4097
Complete: 0
Resume-Tail-SHA256: [SHA256-of-last-128-saved-bytes]

Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 26
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 10000
Body-Format: text/plain
Total-Length: 10000
Content-SHA256: [original-10000-byte-SHA256]
One-To-One: 1
Resume: 1
Resume-Offset: 4097

[128 overlap bytes from original offsets 3969..4096]
[remaining original bytes from offset 4097..9999]
```

Bootstrap payload count is 10000 - 4097 + 128 = 6031. The final /delivery-success contains the full original checksum receipt and is sent on this resumed transfer connection. It is not sent on the earlier status connection.

## 3.11 Completed retry after lost receipt

Return the same persisted receipt with the original completion time; do not duplicate content. A hash-invalid partial is reset; changed immutable delivery metadata conflicts.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /transfer-status
Header-Count: 15
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Transfer-Fingerprint: [64-hex-immutable-fingerprint]
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /transfer-state
Header-Count: 12
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Random: 0123456789ABCDEF
Resume-Offset: 29
Complete: 1
Resume-Tail-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78



[SECOND FRAME ON SAME STATUS CONNECTION]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 17
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Localization: en-US
Country-Of-Origin: IN
Time-Zone: UTC
Send-Time: 1792137660
Random: FEDCBA9876543210
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 394
Content-Type: text/plain; charset=utf-8

Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

## 3.12 Retrieve recipient history

TSV response uses actual TAB and CRLF, with blank CRLF between send-event recipient groups. Requester must have appeared in the responding peer's Sent history. Address shown is documentation-only; not a live contact endpoint.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message-recipients-information-retrieval
Header-Count: 15
Operating-Mode: One-To-One
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Sender's-Login: ASCII-data-alice-account
Sender's-Password: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA ~;~ 0000000000000001 ~;~
    1792137600
Localization: en-US
Country-Of-Origin: IN
Time-Zone: Asia/Kolkata
Send-Time: 1792137600
Random: 0123456789ABCDEF
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600-ABCDEFGHIJ
Delivery-Identifier: 22222222222242228222222222222222
Retrieve-Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message-recipients-information-retrieval
Header-Count: 9
Operating-Mode: One-To-One
Sender-Full-Name: Bob Example
Sender-Email: bob@example.net
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600-ABCDEFGHIJ
Delivery-Identifier: 22222222222242228222222222222222
Retrieve-Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Content-Type: text/tab-separated-values

Bob Example<TAB>bob@example.net<TAB>203.0.113.20<TAB>1792137600
```

# 4. Client/server exchanges - one central server

Alice and Bob use the same server. All central commands below have sender-named credentials, including Bob's download and acknowledgement requests. Account provisioning and peer credential/pin exchange remain out of band; no invented signup endpoint is included.

## 4.1 Alice sends full message

One nested original frame is the entire outer command payload. The server binds sender identity, creates a distinct delivery ID per recipient, archives Sent and returns storage acceptance.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /send-message
Header-Count: 6
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
Request-Identifier: 44444444444444448444444444444444

[NESTED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 20
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>name<TAB>email
11111111111141118111111111111111<TAB>Bob Example<TAB>bob@example.net
```

## 4.2 Bob lists new mail

Listing body is an excerpt of JSON-quoted nonsecret stored headers, with actual CRLF separators. Normal full listing includes the original nonsecret metadata; no credential fields.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-unread-ScurryMails
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-bob-account
Sender's-Password: ASCII-data-bob-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "ScurryMails"
Read: "false"
Category: "0"
Message-Length: "29"
```

## 4.3 Bob retrieves one stored message

Receiver verifies the complete original payload and persists it before submitting a receipt. Use the server-assigned delivery ID from this retrieved copy when constructing the receipt.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-bob-account
Sender's-Password: ASCII-data-bob-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: ScurryMails
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 4.4 Bob submits the delivery receipt

The eight-line body uses original sender identity and frozen time. Server verifies recipient mailbox ownership and checksum before queueing for Alice. It is not acceptance of an invitation or proof of reading.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-acknowledgement
Header-Count: 6
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-bob-account
Sender's-Password: ASCII-data-bob-account-password
Server-Message-Id: 11111111111141118111111111111111

Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
```

## 4.5 Alice fetches delivery receipts

Server removes only returned queue rows after its complete TLS response write. Alice validates against Sent payloads and durably stores the receipt. Crash after server write but before sender storage can lose the fetched receipt; this is not exactly-once consumption.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED RECEIPT FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 18
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 394
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Content-Type: text/plain; charset=utf-8

Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

## 4.6 Subsequent empty receipt fetch

An identical repeated receiver acknowledgement does not recreate a consumed receipt. A digest-only replay ledger may remain.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 0
```

# 5. Client/server exchanges - multiple central servers

Actors: Alice (sender), Server A (sender mailbox), Server B (receiver mailbox), Bob (receiver). The receiver's client moves the original between servers; no direct server-to-server payload exchange is defined. Delegated credentials, binary values and original binding metadata persist unchanged throughout one delivery.

## 5.1 Alice -> Server A: issue token

Full original is archived in Server A Sent. Issuance retry must use unchanged content and random values to receive the original token. The general sender account password is not disclosed to Bob.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /send-message
Header-Count: 11
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
Request-Identifier: 44444444444444448444444444444444
Sender-Host: sender.example.org
Sender-Port: 33338
Token-Request: Binary-data-zero-[256-zero-octets]
Random-Id: Binary-data-00000001-[exactly-256-raw-bytes]
Random-Password: Binary-data-00000002-[exactly-256-raw-bytes]

[NESTED COMPLETE ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 20
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 25
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Sender-Host: sender.example.org
Sender-Port: 33338
Sender's-Token: Binary-data-00000003-[exactly-256-raw-bytes]
Delegated-Login: 0123456789abcdef0123456789abcdef
Delegated-Password: 0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
```

## 5.2 Alice -> Server B: header-only notice

Zero frame payload despite Body-Length/Total-Length describing the referenced original. This acceptance confirms notice storage only; Server B cannot independently authenticate the sender account from remote delegation.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /sender-token-notice
Header-Count: 24
Operating-Mode: Central-Server
Sender's-Login: 0123456789abcdef0123456789abcdef
Sender's-Password: 0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Sender-Host: sender.example.org
Sender-Port: 33338
Random-Id: Binary-data-00000001-[exactly-256-raw-bytes]
Random-Password: Binary-data-00000002-[exactly-256-raw-bytes]
Sender's-Token: Binary-data-00000003-[exactly-256-raw-bytes]
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 7
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
Server-Message-Id: 33333333333343338333333333333333
```

## 5.3 Bob -> Server B: list notices

Listing is abbreviated and redacts delegation secrets; native server also omits binary capabilities from this listing.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-ScurryMails
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-bob-account
Sender's-Password: ASCII-data-bob-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "33333333333343338333333333333333"
Token-Notice: "1"
Subject: "Example subject"
Message-Length: "0"
```

## 5.4 Bob -> Server B: fetch full notice

Only full account-owned retrieval exposes the scoped information needed to contact Server A. Verify source TLS and deployment source-host policy before following a notice-supplied address.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-bob-account
Sender's-Password: ASCII-data-bob-account-password

message-id
33333333333343338333333333333333
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED HEADER-ONLY NOTICE]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /sender-token-notice
Header-Count: 30
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Sender-Host: sender.example.org
Sender-Port: 33338
Random-Id: Binary-data-00000001-[exactly-256-raw-bytes]
Random-Password: Binary-data-00000002-[exactly-256-raw-bytes]
Sender's-Token: Binary-data-00000003-[exactly-256-raw-bytes]
Server-Message-Id: 33333333333343338333333333333333
Token-Notice: 1
Folder: ScurryMails
Read: false
Category: 0
Message-Length: 0
Delegated-Login: 0123456789abcdef0123456789abcdef
Delegated-Password: 0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
```

## 5.5 Bob -> Server A: retrieve original, first chunk

This small example fits one chunk. The nested frame payload is the chunk only; Body-Length, Total-Length and SHA256 describe the full original. Save and verify chunk identity/range before advancing the durable offset.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message-retrieval-from-sender
Header-Count: 26
Operating-Mode: Central-Server
Sender's-Login: 0123456789abcdef0123456789abcdef
Sender's-Password: 0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Sender-Host: sender.example.org
Sender-Port: 33338
Random-Id: Binary-data-00000001-[exactly-256-raw-bytes]
Random-Password: Binary-data-00000002-[exactly-256-raw-bytes]
Sender's-Token: Binary-data-00000003-[exactly-256-raw-bytes]
Resume-Offset: 0
Retrieval-Length: 65536
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED ORIGINAL CHUNK FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 20
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Sender-Host: sender.example.org
Sender-Port: 33338
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 5.6 Bob -> Server A: resumed source retrieval

Separate large-message example with a different originally bound payload. Source chunk resume does not use the peer 128-byte overlap. Reconnect at the last durably saved complete chunk. Reassemble all bytes, verify full SHA256 and descriptors, then import.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message-retrieval-from-sender
Header-Count: 26
Operating-Mode: Central-Server
Sender's-Login: 0123456789abcdef0123456789abcdef
Sender's-Password: 0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 100000
Body-Format: text/plain
Total-Length: 100000
Content-SHA256: [original-100000-byte-sha256]
Sender-Host: sender.example.org
Sender-Port: 33338
Random-Id: Binary-data-00000001-[exactly-256-raw-bytes]
Random-Password: Binary-data-00000002-[exactly-256-raw-bytes]
Sender's-Token: Binary-data-00000003-[exactly-256-raw-bytes]
Resume-Offset: 65536
Retrieval-Length: 65536
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED ORIGINAL CHUNK FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 20
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 100000
Body-Format: text/plain
Total-Length: 100000
Content-SHA256: [original-100000-byte-sha256]
Sender-Host: sender.example.org
Sender-Port: 33338
Resume-Offset: 65536

Binary-data-[remaining-34464-original-bytes]
```

## 5.7 Bob -> Server B: import original for storage

Receiver credentials authorize this operation. Neither outer nor nested frame includes Token-Request, Sender's-Token, Random-Id, Random-Password or delegated secrets. Preserve original sender, subject, endpoint, body and attachment descriptors. Replace notice while preserving mailbox ID; retries must not duplicate mail.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message-retrieval-from-sender-for-storage
Header-Count: 6
Operating-Mode: Central-Server
Receiver's-Login: ASCII-data-bob-account
Receiver's-Password: ASCII-data-bob-account-password
Notice-Id: 33333333333343338333333333333333

[NESTED COMPLETE VERIFIED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 22
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
One-To-One: 1
Resume: 0
Resume-Offset: 0
Sender-Host: sender.example.org
Sender-Port: 33338

UTF-data-00000001-Hello Bob!
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 7
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
Server-Message-Id: 33333333333343338333333333333333
```

## 5.8 Bob -> Server A: submit receipt

Submit only after verified local persistence and successful import. Persist an encrypted pending journal before import/ack work; restart can finish idempotent import and resend the exact original receipt. Scoped capability validation is not an independent receiver signature.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-acknowledgement
Header-Count: 24
Operating-Mode: Central-Server
Sender's-Login: 0123456789abcdef0123456789abcdef
Sender's-Password: 0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Sender-Host: sender.example.org
Sender-Port: 33338
Random-Id: Binary-data-00000001-[exactly-256-raw-bytes]
Random-Password: Binary-data-00000002-[exactly-256-raw-bytes]
Sender's-Token: Binary-data-00000003-[exactly-256-raw-bytes]

Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
```

## 5.9 Alice -> Server A: retrieve receipts

Consume fetched acknowledgement rows after successful complete write. Empty next poll and duplicate submission behavior match section 4.6.

Sender credentials mean scoped values in steps 5.2, 5.5, 5.6 and 5.8; Bob account values in 5.3/5.4; receiver-named Bob values only in 5.7.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

### RESPONSE

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED RECEIPT FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delivery-success
Header-Count: 18
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 394
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Content-Type: text/plain; charset=utf-8

Checksum-Algorithm: SCURRY-PRNG-SHA512-1
Message-SHA512: 1c250f9cfdd992f19da748bfc1ed922ba712db50ba2e9f36100ec36ee6fe8e5fbe9a048a0186bbfc5bf6e2cb8acdbe
    6d9a7aeb873328a9d4a535398b6eceacdf
Sender-Name: Alice Example
Receiver-Name: Bob Example
Sender-Email: alice@example.org
Receiver-Email: bob@example.net
Sent-Time: 2026-10-16T08:00:00+00:00 [UTC]
Received-Time: 2026-10-16T08:01:00+00:00 [UTC]
```

# 6. Client/contact-book or address-book exchanges

The bundled companion directory listens on loopback UDP 33336 by default. One UTF-8 JSON object per datagram, maximum 8192 bytes. The ASCII JSON below is the actual datagram shape, not HTTP syntax. Source code still contains some legacy ScurryMsg.com directory branding. No SCURRY Protocol header is sent to the directory.

## 6.1 Directory availability

Transport: client UDP datagram -> local directory, then directory response -> client source port. Square-bracket IP values are placeholders; actual imported contacts require valid public IPs.

**REQUEST**

```
{
  "version": 1,
  "op": "ping",
  "request_id": "check-1"
}
```

**RESPONSE**

```
{
  "version": 1,
  "request_id": "check-1",
  "ok": true,
  "app": "ScurryMsg.com"
}
```

## 6.2 Exact name/email lookup with optional city

Transport: client UDP datagram -> local directory, then directory response -> client source port. Square-bracket IP values are placeholders; actual imported contacts require valid public IPs.

**REQUEST**

```
{
  "version": 1,
  "op": "get",
  "request_id": "find-1",
  "name": "Bob Example",
  "email": "bob@example.net",
  "city": "Bengaluru"
}
```

**RESPONSE**

```
{
  "version": 1,
  "request_id": "find-1",
  "ok": true,
  "contact": {
    "id": "88888888-8888-4888-8888-888888888888",
    "name": "Bob Example",
    "email": "bob@example.net",
    "city": "Bengaluru",
    "address": "[public-IPv4-address]",
    "ip_version": 4,
    "imported_at": 1792137600,
    "expires_at": null,
    "expired": false,
    "freshness": "unknown"
  }
}
```

## 6.3 Lookup by local contact ID

Transport: client UDP datagram -> local directory, then directory response -> client source port. Square-bracket IP values are placeholders; actual imported contacts require valid public IPs.

**REQUEST**


```
{
  "version": 1,
  "op": "get",
  "request_id": "find-2",
  "id": "88888888-8888-4888-8888-888888888888"
}
```


**RESPONSE**


```
{
  "version": 1,
  "request_id": "find-2",
  "ok": true,
  "contact": {
    "id": "88888888-8888-4888-8888-888888888888",
    "name": "Bob Example",
    "email": "bob@example.net",
    "city": "Bengaluru",
    "address": "[public-IPv4-address]",
    "ip_version": 4,
    "imported_at": 1792137600,
    "expires_at": null,
    "expired": false,
    "freshness": "unknown"
  }
}
```


## 6.4 First list page

Transport: client UDP datagram -> local directory, then directory response -> client source port. Square-bracket IP values are placeholders; actual imported contacts require valid public IPs.

**REQUEST**


```
{
  "version": 1,
  "op": "list",
  "request_id": "page-1",
  "offset": 0,
  "limit": 1
}
```


**RESPONSE**


```
{
  "version": 1,
  "request_id": "page-1",
  "ok": true,
  "contacts": [
    {
      "id": "88888888-8888-4888-8888-888888888888",
      "name": "Bob Example",
      "email": "bob@example.net",
      "city": "Bengaluru",
      "address": "[public-IPv4-address]",
      "ip_version": 4,
      "imported_at": 1792137600,
      "expires_at": null,
      "expired": false,
      "freshness": "unknown"
    }
  ],
  "total": 2,
  "next_offset": 1
}
```

## 6.5 Next list page / IPv6 contact

Transport: client UDP datagram -> local directory, then directory response -> client source port. Square-bracket IP values are placeholders; actual imported contacts require valid public IPs.

**REQUEST**

```
{
  "version": 1,
  "op": "list",
  "request_id": "page-2",
  "offset": 1,
  "limit": 1
}
```

**RESPONSE**

```
{
  "version": 1,
  "request_id": "page-2",
  "ok": true,
  "contacts": [
    {
      "id": "99999999-9999-4999-8999-999999999999",
      "name": "Bob Example",
      "email": "bob@example.net",
      "city": "Bengaluru",
      "address": "[public-IPv6-address]",
      "ip_version": 6,
      "imported_at": 1792137600,
      "expires_at": null,
      "expired": false,
      "freshness": "unknown"
    }
  ],
  "total": 2,
  "next_offset": null
}
```

## 6.6 Protected directory lookup

When a listener requires a token, send it on every operation, including ping/list. Plain UDP does not encrypt the token. Default loopback listeners need no token.

**REQUEST**

```
{
  "version": 1,
  "op": "get",
  "request_id": "protected-1",
  "token": "ASCII-data-shared-directory-token",
  "id": "88888888-8888-4888-8888-888888888888"
}
```

**RESPONSE**


```
{
  "version": 1,
  "request_id": "protected-1",
  "ok": true,
  "contact": {
    "id": "88888888-8888-4888-8888-888888888888",
    "name": "Bob Example",
    "email": "bob@example.net",
    "city": "Bengaluru",
    "address": "[public-IPv4-address]",
    "ip_version": 4,
    "imported_at": 1792137600,
    "expires_at": null,
    "expired": false,
    "freshness": "unknown"
  }
}
```


## 6.7 not_found

No matching local contact.

**REQUEST**


```
{
  "version": 1,
  "op": "get",
  "name": "Missing Person",
  "email": "missing@example.net",
  "request_id": "error-not_found"
}
```


**RESPONSE**


```
{
  "version": 1,
  "request_id": "error-not_found",
  "ok": false,
  "error": "not_found"
}
```


## 6.7 ambiguous

More than one city matches; retry with city or local ID.

**REQUEST**


```
{
  "version": 1,
  "op": "get",
  "name": "Bob Example",
  "email": "bob@example.net",
  "request_id": "error-ambiguous"
}
```


**RESPONSE**


```
{
  "version": 1,
  "request_id": "error-ambiguous",
  "ok": false,
  "error": "ambiguous",
  "message": "Supply city or contact id."
}
```


## 6.7 expired

Contact has an explicit past expiry. Manually obtain/import a fresh .conf.

**REQUEST**

```
{
  "version": 1,
  "op": "get",
  "id": "88888888-8888-4888-8888-888888888888",
  "request_id": "error-expired"
}
```

**RESPONSE**

```
{
  "version": 1,
  "request_id": "error-expired",
  "ok": false,
  "error": "expired"
}
```

## 6.7 unauthorized

Protected listener rejects a missing or mismatching token.

**REQUEST**

```
{
  "version": 1,
  "op": "ping",
  "token": "wrong-token",
  "request_id": "error-unauthorized"
}
```

**RESPONSE**

```
{
  "version": 1,
  "request_id": "error-unauthorized",
  "ok": false,
  "error": "unauthorized"
}
```

## 6.7 bad_request

Unsupported operation, bad version/type, malformed JSON/UTF-8, packet too large or invalid pagination can produce bad_request.

**REQUEST**

```
{
  "version": 1,
  "op": "delete",
  "id": "88888888-8888-4888-8888-888888888888",
  "request_id": "error-bad_request"
}
```

**RESPONSE**

```
{
  "version": 1,
  "request_id": "error-bad_request",
  "ok": false,
  "error": "bad_request"
}
```

## 6.8 Loss, timeout and correlation

No datagram is an acknowledgement until request_id matches. UDP can lose or reorder packets: retry with bounded timeout and reject an unrelated response. The bundled lookup helper makes up to three attempts with a two-second timeout. No reply means timeout, not an invented error datagram. The address book supplies the address only; peer contact settings supply TCP port, TLS pin and pair credentials.

## 6.9 Add, update, delete and export contacts

These are local GUI/file operations, not network endpoints. Add/update imports a .conf; updates retain the local ID. Delete removes only the local contact; export writes .conf. There is no UDP add/update/delete exchange to show. Unknown fields/PINs are discarded. Optional City/ExpiresAt are extensions; expiry is unknown if absent.

```
[ScurryLink]
FullName="Bob Example"
Email="bob@example.net"
Address=[public-IPv4-or-IPv6-address]
City="Bengaluru"
ExpiresAt=1792224000
```

## 6.10 Optional user-triggered HTTPS contact download

This is the one genuine HTTP example: a configured city backend, not SCURRY 1.2 or the local UDP API. The companion client supports the following POST and parses the returned .conf; an operational city backend must exist. No automatic public IP update or scheduled lookup is implied. Host and Content-Length are ordinary HTTP transport headers, not SCURRY headers. Length below is a computed-by-HTTP-client placeholder.

### Download from configured city backend

Client enforces HTTPS origin, rejects redirects, bounds .conf to 16 KiB, validates returned name/email and never submits a PIN. Response media-type/status shown is an illustrative compatible backend response, not a newly deployed service. An HTTP error is reported generically; optional JSON detail may be displayed.

**REQUEST**

```
POST /api/search HTTP/1.1
Host: city.example.org
Content-Type: application/json
Accept: text/plain
Content-Length: [JSON-byte-count]

{"name": "Bob Example", "email": "bob@example.net", "city": "Bengaluru", "address": "", "pin": ""}
```

**RESPONSE**

```
HTTP/1.1 200 OK
Content-Type: text/plain; charset=utf-8
Content-Length: [conf-byte-count]

[ScurryLink]
FullName="Bob Example"
Email="bob@example.net"
Address=[public-IP-address]
```

# 7. Exhaustive central endpoint and filter catalog

The following catalog covers every currently implemented literal endpoint and every folder/filter expansion using Work as the example folder. Work is not the only possible folder. Category variants enumerate 1..8. Read/unread/category suffixes require a folder token. Replace folder names with URI-encoded UTF-8 tokens (max 192 encoded characters and 64 decoded scalars); reject controls, raw slash, dot/dot-dot and malformed escapes. Protocol slot length still limits the complete path.

Commands in this section use Alice's account for illustration; listing/download/delete normally use the mailbox owner's values. Credential pairs always identify the acting account. Source retrieval, notice and import have full required-field examples in section 5; catalog cross-references avoid misleading empty requests.

| Endpoint family | Request body / result |
|---|---|
| /sender-token-notice | Full request and response: section 5.2 |
| /message-retrieval-from-sender | Full request and response: section 5.5 and 5.6 |
| /message-retrieval-from-sender-for-storage | Full request and response: section 5.7 |
| /delivery-acknowledgement | Full request and response: section 4.4 and 5.8 |
| /delivery-success | Full request and response: section 4.5/4.6 and 5.9 |

## 7.1 /list-functionality

Metadata object abbreviated; query the actual service for enabled functions. It is informational, not version negotiation.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-functionality
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/json; charset=utf-8
Message-Count: 0

{
  "protocol_version": "1.2",
  "operating_mode": "Central-Server",
  "schema": "scurry.server-functionality/1",
  "server": {
    "id": "ASCII-data-service-id",
    "name": "Central mailbox",
    "software": "ScurryServer",
    "software_version": "1.10.0"
  },
  "host": "[hostnames-ip_addresses-reverse_dns-lookup_status-object]",
  "generated_utc": "2026-10-23T00:00:00Z",
  "functionalities": [
    {
      "path": "/list-functionality",
      "kind": "literal",
      "authentication_required": true
    }
  ],
  "response_functionalities": [
    "/central-response"
  ]
}
```

## 7.7 /search-all-messages

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /search-all-messages
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

{"subject": "Example", "sender": "alice@", "text": "Hello"}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "ScurryMails"
Read: "false"
Category: "0"
```

## 7.8 /get-all-messages-with-search-criteria

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-with-search-criteria
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

{"subject": "Example"}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777477787777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: ScurryMails
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.9 /get-one-message-with-search-criteria

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-one-message-with-search-criteria
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

{"subject": "Example"}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: ScurryMails
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.10 /send-message

Acceptance only. Server-scoped directory expansion creates independent addressed deliveries and Sent copies; receivers later submit independent receipts. Retry unchanged Request-Identifier and payload for idempotency.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /send-message
Header-Count: 6
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
Request-Identifier: 44444444444444448444444444444444

[NESTED COMPLETE ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 20
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>name<TAB>email
11111111111141118111111111111111<TAB>Bob Example<TAB>bob@example.net
```

## 7.11 /send-message-with-search-criteria

Acceptance only. Server-scoped directory expansion creates independent addressed deliveries and Sent copies; receivers later submit independent receipts. Retry unchanged Request-Identifier and payload for idempotency.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /send-message-with-search-criteria
Header-Count: 7
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
Request-Identifier: 44444444444444448444444444444444
Additional-Recipients: text/tab-separated-values ~;~ UTF-8 ~;~ none ~;~ [tsv-byte-count] ~;~
    [nested-frame-and-payload-byte-count] ~;~ recipients.tsv

[NESTED COMPLETE ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 20
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 22222222222242228222222222222222
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!

[SEPARATE SELECTOR TSV ATTACHMENT]
match<TAB>name<TAB>email
exact<TAB>Bob Example<TAB>bob@example.net
wildcard<TAB><TAB>*@example.org
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>name<TAB>email
11111111111141118111111111111111<TAB>Bob Example<TAB>bob@example.net
```

## 7.12 /set-status

Status property update, not an automatic outbound message. Unknown JSON fields and overlong values are rejected.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /set-status
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

{"value": "Available"}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
```

## 7.13 /set-status-message

Status property update, not an automatic outbound message. Unknown JSON fields and overlong values are rejected.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /set-status-message
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

{"value": "UTF-data-status-text"}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
```

## 7.14 /set-holiday-message

Status property update, not an automatic outbound message. Unknown JSON fields and overlong values are rejected.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /set-holiday-message
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

{"value": "UTF-data-status-text"}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
```

## 7.15 /set-grouped-status

Status property update, not an automatic outbound message. Unknown JSON fields and overlong values are rejected.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /set-grouped-status
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

{"group": "Work", "value": "Away"}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
```

## 7.16 /set-grouped-status-message

Status property update, not an automatic outbound message. Unknown JSON fields and overlong values are rejected.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /set-grouped-status-message
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

{"group": "Work", "value": "Away"}
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/octet-stream
Message-Count: 0
```

## 7.17 /get-all-,messages

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-,messages
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777477787777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: ScurryMails
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.18 /list-all-messages

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "ScurryMails"
Read: "false"
Category: "0"
```

## 7.19 /list-all-messages-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "0"
```

## 7.20 /list-all-messages-read-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-read-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "true"
Category: "0"
```

## 7.21 /list-all-messages-unread-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-unread-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "0"
```

## 7.22 /list-all-messages-category1-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-category1-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "1"
```

## 7.23 /list-all-messages-category2-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-category2-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "2"
```

## 7.24 /list-all-messages-category3-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-category3-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "3"
```

## 7.25 /list-all-messages-category4-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-category4-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "4"
```

## 7.26 /list-all-messages-category5-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-category5-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "5"
```

## 7.27 /list-all-messages-category6-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-category6-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "6"
```

## 7.28 /list-all-messages-category7-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-category7-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "7"
```

## 7.29 /list-all-messages-category8-Work

Listing excerpt; values are ASCII JSON strings. Empty result uses Message-Count: 0 and no body.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /list-all-messages-category8-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/scurry-headers; charset=us-ascii
Message-Count: 1

Server-Message-Id: "11111111111141118111111111111111"
Subject: "Example subject"
Folder: "Work"
Read: "false"
Category: "8"
```

## 7.30 /get-all-messages

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: ScurryMails
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.31 /get-all-messages-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.32 /get-all-messages-read-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-read-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: true
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.33 /get-all-messages-unread-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-unread-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.34 /get-all-messages-category1-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-category1-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 1
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.35 /get-all-messages-category2-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-category2-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 2
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.36 /get-all-messages-category3-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-category3-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 3
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.37 /get-all-messages-category4-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-category4-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 4
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.38 /get-all-messages-category5-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-category5-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 5
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.39 /get-all-messages-category6-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-category6-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 6
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.40 /get-all-messages-category7-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-category7-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777477778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 7
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.41 /get-all-messages-category8-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-all-messages-category8-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777477778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 8
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.42 /get-message

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: ScurryMails
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.43 /get-message-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.44 /get-message-read-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-read-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: true
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.45 /get-message-unread-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-unread-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777477778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 0
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.46 /get-message-category1-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-category1-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 1
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.47 /get-message-category2-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-category2-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 2
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.48 /get-message-category3-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-category3-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 3
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.49 /get-message-category4-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-category4-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 4
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.50 /get-message-category5-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-category5-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777477778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 5
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.51 /get-message-category6-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-category6-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 6
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.52 /get-message-category7-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-category7-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777747778777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 7
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.53 /get-message-category8-Work

Every selected ID must belong to the account and match route folder/filter. Unfiltered get-all also accepts an empty body to select all; get-message requires exactly one ID.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /get-message-category8-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: application/x-scurry-frames
Message-Count: 1

[NESTED COMPLETE STORED ORIGINAL FRAME]
Protocol: SCURRY 1.2 UTF-8 UTF-8 /message
Header-Count: 27
Operating-Mode: Central-Server
Sender-Full-Name: Alice Example
Sender-Email: alice@example.org
Recipient-Name: Bob Example
Recipient-Email: bob@example.net
Subject: Example subject
Message-Identifier: AliceExample ~;~ alice@example.org ~;~ fbf7c6aec1d4280b7c2704c1c0478bd6 ~;~
    16bac85075f74ef26e34f020702d4d74 ~;~ 55555555-5555-4555-8555-555555555555 ~;~ 1792137600
Delivery-Identifier: 77777777777477787777777777777777
Send-Time: 1792137600
Message-Sent-Time: 1792137600
Receipt-Algorithm: SCURRY-PRNG-SHA512-1
Body-Length: 29
Body-Format: text/plain
Total-Length: 29
Content-SHA256: 419ec15916c41de52ce2fd32e7407bb8c502af5671bb668b4bfbb6b0d54f1a78
Server-Message-Id: 11111111111141118111111111111111
Sender-Account: ASCII-data-alice-account-key
Folder: Work
Read: false
Category: 8
Message-Length: 29
Additional-Recipients-For-Message: Bob Example<TAB>bob@example.net
One-To-One: 1
Resume: 0
Resume-Offset: 0

UTF-data-00000001-Hello Bob!
```

## 7.54 /delete-message

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.55 /delete-message-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.56 /delete-message-read-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-read-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.57 /delete-message-unread-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-unread-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.58 /delete-message-category1-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-category1-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.59 /delete-message-category2-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-category2-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.60 /delete-message-category3-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-category3-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.61 /delete-message-category4-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-category4-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.62 /delete-message-category5-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-category5-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.63 /delete-message-category6-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-category6-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.64 /delete-message-category7-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-category7-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7.65 /delete-message-category8-Work

Nonempty explicit selection only, all-or-nothing within the account. Missing/foreign/out-of-route IDs fail; never omit body to delete all.

**REQUEST**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /delete-message-category8-Work
Header-Count: 5
Operating-Mode: Central-Server
Sender's-Login: ASCII-data-alice-account
Sender's-Password: ASCII-data-alice-account-password

message-id
11111111111141118111111111111111
```

**RESPONSE**

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 200
Content-Type: text/tab-separated-values; charset=utf-8
Message-Count: 1

message-id<TAB>result
11111111111141118111111111111111<TAB>deleted
```

## 7. Errors and rejected exchanges

Malformed request, selection, descriptor, conflicting idempotency content or range.

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 400
Content-Type: application/octet-stream
Message-Count: 0
```

Authentication/authorization, ownership or delegated capability mismatch.

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 403
Content-Type: application/octet-stream
Message-Count: 0
```

Unavailable account-scoped selection; do not disclose another account.

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 404
Content-Type: application/octet-stream
Message-Count: 0
```

Configured host has no implementation for the callback.

```
Protocol: SCURRY 1.2 UTF-8 UTF-8 /central-response
Header-Count: 6
Operating-Mode: Central-Server
Status-Code: 501
Content-Type: application/octet-stream
Message-Count: 0
```

These are generic host response shapes, not a promise every failure gets a reply. Framing/TLS failures, invalid version, misplaced Protocol, malformed Unicode or bad authentication may close the connection. Peer failures leave Outbox pending and do not fabricate a success receipt. Never echo account passwords or

binary secrets in an error.

# 8. Coverage, limitations and package contents

This document expands 65 unique central paths/templates (including five cross-referenced 1.2 operations) plus all seven peer content paths, status/resume/replay, capabilities, history and the contact directory operations/errors. Folders, attachment index combinations, JSON text and recipients remain parameterized. No new account-registration, direct S2S, HTTP relay/push, remote contact mutation, or unapproved 1.1 resource endpoint is implied.

Security/deployment limits remain as recorded in the design: destructive-fetch crash window, bearer delegated receipt scope, and open token expiry/revocation, notice-abuse and source-host policy requirements. A checksum is not a digital signature. This documentation pass cross-checks source contracts but does not replace platform/runtime acceptance testing.

Included under docs/: illustrated design PDF, design Markdown with editable Mermaid diagrams, this exchange-reference PDF and Markdown, earlier per-endpoint message examples, current RFC and validation reports. The source release remains 1.10.0; this is a documentation packaging update.